YouTube-DNN-RecSys — Candidate Generation Architecture

Data processing, storage, dnn.py model, prediction services, and support utilities

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

PROJECT OVERVIEW

YouTube-DNN-RecSys is a TensorFlow and Spark implementation of the candidate-generation stage of YouTube recommendations.
The architecture comprises data processing, storage, the dnn.py model, prediction services, and support utilities.

DATA PROCESSING · offline sample build

Python Pipeline

data2tfrecords.py
random training data

TFRecords
train / evaluation

read_tfrecords.py
dataset reader

Spark Pipeline

MakeDataOne.scala
distributed generation

MakeDataTwo.scala
TFRecords output

HDFS
dataset storage

STORAGE · datasets and checkpoints

data/tfrecords/
train/train.tfrecords
evaluation/evaluation.tfrecords

data/checkpoints/
ckpt/checkpoint
ckpt/eval · modelpath/

MODEL TRAINING · dnn.py

Input
visit_items_index × 5
continuous_features × 16

Item Embedding
500 items × 16

Average Pooling
history vector

DNN Tower
Dense 64 + ReLU
Dropout · Dense 32 + ReLU
user vector 16

Softmax Output
500 classes
cross entropy · Top-K

PREDICTION · serving paths

PredictUserVector.scala
SavedModel
user vector head

ItemEmbeddingPredictor.scala
item embedding table
FeatureBuilder

load_dnn_model.py
SavedModel load
online / batch inference

UTILITIES · training support

tensor_board.py
TensorBoard

feature_column.py
feature spec

self_defined_network_layer.py
custom TF layers

Spark Base
BaseSparkLocal.scala
BaseSparkOnline.scala

Configuration
vars.prod.properties
vars.sit.properties

The model contains input, item embedding, average pooling, the user tower, and softmax. Python and Spark each provide data and prediction components.